NNN-C-1434C
January 21, 1977
SUPERSEDING
Fed. Spec. NNN-C-1434B
August 29, 1972

FEDERAL SPECIFICATION

COVER GLASS, MICROSCOPE SLIDE

This specification was approved by the Commissioner, Federal Supply Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for microscope slide cover glasses for microscopic observation.

1.2 Classification.

1.2.1 Types, sizes and thicknesses. The cover glasses shall be of the following types, sizes and thicknesses (see 6.1):

Type I--For blood counting chambers

Size 1--20 millimeters (mm.) by 26 mm. by 0.4 mm. thick.

Type II--For microscopy

Size 1--Square, 18 mm. by 18 mm.
Size 2--Square, 22 mm. by 22 mm.
Size 3--Square, 24 mm. by 24 mm.
Size 4--Square, 25 mm. by 25 mm.
Size 5--Rectangle, 22 mm. by 30 mm.
Size 6--Rectangle, 22 mm. by 40 mm.
Size 7--Rectangle, 22 mm. by 50 mm.
Size 8--Round, 18 mm. in diameter.
Size 9--Round, 25 mm. in diameter.
Size 10--Rectangle, 24 mm. by 40 mm.
Size 11--Rectangle, 20 mm. by 40 mm.
Size 12--Rectangle, 24 mm. by 50 mm.

Thickness No. 0--0.085 mm. to 0.13 mm.
Thickness No. 1--0.13 mm. to 0.17 mm.
Thickness No. 1-1/2--0.16 mm. to 0.19 mm.
Thickness No. 2 --0.17 mm. to 0.25 mm.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:49Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:49Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standard:

C162 - Standard Definitions of Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Bureau of Standards Note No. 579:

Testing of Cover Glasses for Hemacytometer Chambers.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Material. Glass shall be transparent and colorless.

3.2 Dimensions.

3.2.1 Thickness. Minimum and maximum thicknesses shall be as indicated in 1.2.1. The thickness of a single type I cover glass shall not vary by more than 0.06 mm.; and the thickness of a single type II cover glass shall not vary by more than 0.01 mm. when tested as specified in 4.4.1.

3.2.2 Size. Side lengths and diameters shall be as indicated in 1.2.1. Tolerances on each of these dimensions shall be ± 1 mm.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:49Z
Check the source to verify that this is the current version before use.

3.3 Planeness. The planeness of type I cover glasses shall not deviate from a true plane by more than 0.002 mm. when tested as specified in 4.4.2.

3.4 Performance.

3.4.1 Solubility. No pink or red color shall be visible along the edges of or through the superimposed types I and II cover glasses when tested as specified in 4.4.3.

3.4.2 Flexibility. Except for thickness no. 2, type II cover glasses shall be capable of conforming to the mandrel when tested as specified in 4.4.4.

3.4.3 Warpage. Type II cover glasses shall not compress more than shown in table I when tested as specified in 4.4.5.

TABLE I. Compression

Thickness no.	Compression in mm. (Warpage)
0	2.13
1	1.42
1-1/2	1.19
2	1.01

3.5 Workmanship. Workmanship shall be first class throughout. When viewed with a good light against a dark background, cover glasses shall be free of defects listed in table IV and any other defects which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:49Z
Check the source to verify that this is the currentversion before use.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. Unit of product shall be one cover glass.

TABLE II. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. When cover glass is unit of product, not more than 10 cover glasses shall be taken from one package.

TABLE III. For tests

Unit of product	Characteristic	Requirement	Test procedure	Inspection level	AQL
Cover glass	Thickness	3.2.1	4.4.1	S-4	.65
Cover glass (type I)	Planeness	3.3	4.4.2	S-3	1.0
Cover glass	Solubility	3.4.1	4.4.3	S-3	*
Cover glass (type II)	Flexibility	3.4.2	4.4.4	S-3	4.0**
1 oz. package (type II)	Warpage	3.4.3	4.4.5	S-1	*

* Acceptance number shall be zero. Extra glasses shall be sampled where necessary to provide specified quantities.

** In the event the number of defectives in a particular lot exceeds the acceptance number for flexibility testing, additional sampling may be conducted to determine acceptability of the lot. The lot

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:49Z
Check the source to verify that this is the current version before use.

shall be sampled an additional four times. For each additional sampling, the inspection level shall be the same as it was for the original sampling. The number of defectives found in each of the five groups of testing samples (i.e., including those determined in the original testing) shall be averaged. If the average number of defectives does not exceed the original acceptance number, the lot shall be considered acceptable.

4.3 Examination. Cover glasses shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in table IV.

TABLE IV. Classification of defects*

Category:	Defects
Major:	
101	Cover glass not clear or not free of fogginess.
102	Cover glass not free of cracks.
103	Cover glass not free of chips or breaks.
104	Cover glass not free of sharp edges or burrs.
105	Cover glass not free of jagged edges.
106	Cover glass not free of bubbles and blisters.
107	Cover glass not free of stones and seeds.
108	Cover glass not free of checks and crizzles.
109	Cover glass not free of warping.
110	Cover glass not free of grease, dirt or other foreign matter.
111	Cover glass not free of oil, oily-type substance or stain.
Minor:	
201	Cover glass not free of striae.
202	Cover glass not free of scratches.

* Terms are as defined in ASTM designation C162.

4.3.1 Dimensional examination. The cover glasses shall be examined for defects in dimensions (except thickness). Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:49Z
Check the source to verify that this is the current version before use.

4.4.1 Thickness. Thickness shall be measured with a standard cover glass gauge or micrometer thickness gauge.

4.4.2 Planeness. Planeness shall be determined in accordance with NBS *Technical Note No. 579.*

4.4.3 Solubility. Twenty four cover glasses shall be cleaned by immersion in distilled water for one minute, and removed and air-dried for one minute in a vertical position. This cleaning operation shall be performed three times, using a new quantity of distilled water for each immersion. Add 300 milliliters (ml.) of distilled water to an alkali-resistant Erlenmeyer flask of suitable size and boil for ten minutes. Allow water to cool for three minutes and then decant until the volume in the flask is approximately 100 ml. Keeping the flask covered, let it stand until the water has cooled to approximately 70° Centigrade (°C.). Add 0.2 ml. of U.S.P. standard phenolphthalein solution and continue to cool to 60°C. Add the cover glasses, one at a time. The cover glasses shall be superimposed by tilting and gently swirling the flask. The flask shall be placed on a white background. This may be achieved by putting a white piece of paper under the flask. Let the solution cool toward room temperature for one hour without agitating the cover glasses. View the cover glasses through the neck of the erlenmeyer flask. Absorption of carbon dioxide from the atmosphere must be avoided during this test.

4.4.4 Flexibility. Cover glass shall be placed on a mandrel which has the diameter specified in table V. The longest dimension of the cover glass shall be at right angles to the axis of the mandrel. The cover glass shall then be gently bent to conform to the mandrel, with its center resting on the mandrel and both ends pressed down simultaneously. Pressure should be exerted as evenly as possible over the entire cover glass. Each cover glass shall be bent one time only.

TABLE V. Mandrel diameters

Thickness No.	Mandrel diameter in inches
0	4
1	4-1/2
1-1/2	5-1/4
2	Not required

4.4.5 Warpage. The number of cover glasses specified in table VI shall be stacked on a flat surface. The height of the stack shall be measured to within ± 0.05 mm. Place a thin, stiff aluminum plate cut to the size of the cover glasses on top of the stack. Place a 100-gram weight on top of the plate. Again measure the height of the stack to ± 0.05 mm. The difference between the unloaded and the loaded readings is the compression described in 3.4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:49Z
Check the source to verify that this is the current version before use.

TABLE VI. Quantity of cover glasses for warpage test

Thickness No.	Quantity of cover glasses
0	140
1	100
1-1/2	86
2	70

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table VII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

TABLE VII. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components or containers distorted.

4.5.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-585, PPP-B-601, PPP-B-621 or PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package.

5.1.1.1.1 Type 1. Individual cover glasses shall be wrapped in soft

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:49Z
Check the source to verify that this is the current version before use.

paper or a pair of cover glasses, separated by soft paper, may be wrapped in paper or other suitable material. Twelve wrapped, or separated and wrapped, cover glasses shall be unit packaged in a suitable box. Packages shall contain cover glasses of one size only.

5.1.1.1.2 Type II. Unit package of cover glasses, weighing a minimum of one ounce, shall be packaged in a plastic container of suitable size with a hinged lid and snap-type fastener. As an alternate to the hinged container, unit package shall be a container of suitable size with a molded plastic bottom section and a top fabricated of plastic, fiberboard, or heavy chipboard. Each unit package (all types of containers specified above) shall be in an upright position permitting ready dispensing when the container is opened. Interior of each unit package shall be cushioned with lint-free material to insure protection of cover glasses and to insure against spilling of cover glasses when container is opened. Container closure shall be adequate to prevent spilling of contents under normal handling.

5.1.1.2 Intermediate package. Unless otherwise specified, twelve unit packages or ten unit packages shall be packaged in a snug-fitting box constructed in accordance with PPP-B-676. When hinged containers (see 5.1.1.1.2) are furnished, dividers shall be furnished to prevent contact of hinges and hasps and adjacent containers. As an alternate to the use of dividers, each hinged container shall be completely wrapped in kraft paper of suitable weight. When containers specified as an alternate to the hinged type are furnished, protective dividers or wraps shall not be required. Wrapping shall be properly secured. Intermediate package closure shall be adequate to prevent spilling of contents under normal handling.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Unless otherwise specified, ten intermediate packages of type I or thirty six intermediate packages of type II shall be packed in an exterior container, designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseastype; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to the special requirements as specified in PPP-B-636. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:49Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Unless otherwise specified, thirty six intermediate packages shall be packed in an exterior container, designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Fiberboard boxes shall conform to the special requirements as specified in PPP-B-636. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Size and thickness required (see 1.2.1).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Type I, size 2 of NNN-C-1434B has been deleted from this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:49Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
HEW - NIH
VA - DMS

PREPARING ACTIVITY: DSA - DM

Project No. 6640-0965

U. S. GOVERNMENT PRINTING OFFICE : 1977 - 241-233/1009

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NNN-C-1434C - COVER GLASS, MICROSCOPE SLIDE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:49Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:49Z
Check the source to verify that this is the current version before use.